

Question Paper

Exam Date & Time: 01-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - JANUARY 2026

Circuits and Systems [ECM_2123]

Marks: 50

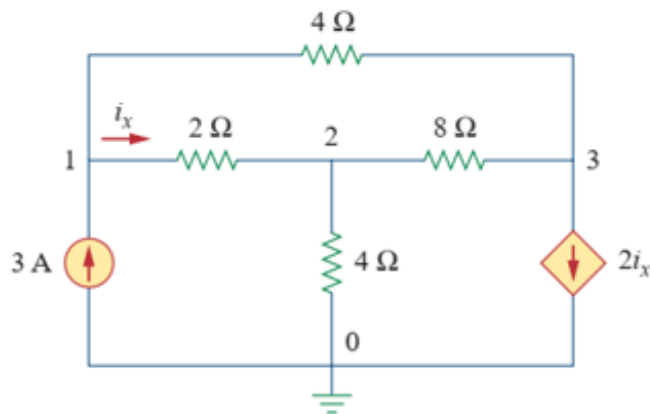
Duration: 180 mins.

PART-A

Answer all the questions.

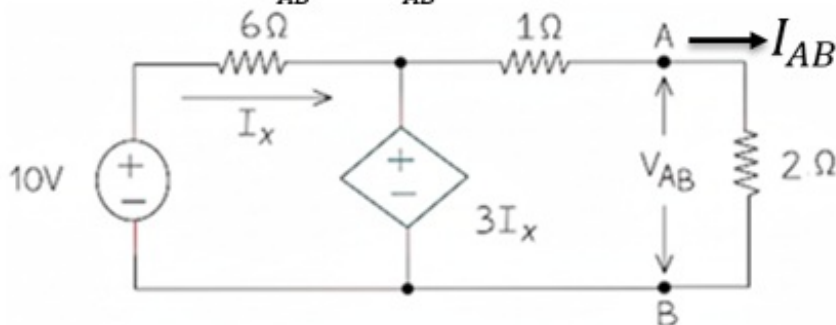
- 1) Determine the voltages at the nodes in the given circuit. (4)

A)



Course Outcome: "CO1" / Cognitive Level: "Application"

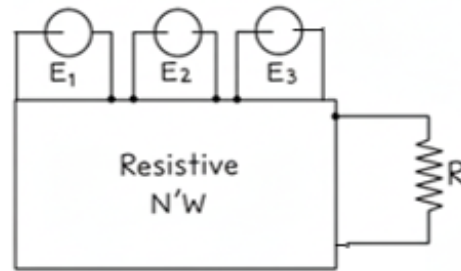
- B) Find the value of V_{AB} and I_{AB} using Thevenin's Theorem. (4)



Course Outcome: "CO1" / Cognitive Level: "Application"

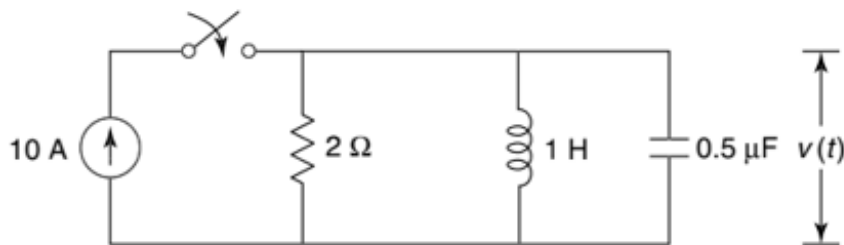
- C) (2)

In the circuit shown in the figure, the power consumed in the resistance R is measured when **one source is acting at a time**. The three measured values of power are **18 W, 50 W, and 98 W**. When the three sources are acting **simultaneously**. Determine the possible **maximum and minimum** values of power in R .



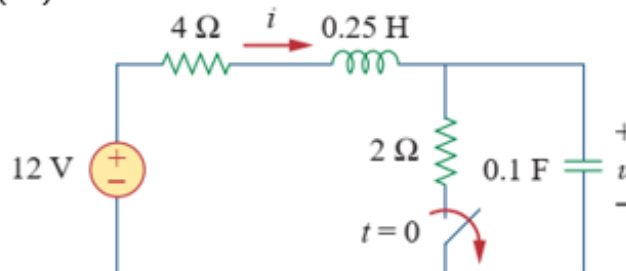
Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) For the network shown in Figure, the switch is closed at $t = 0$, determine v , dv/dt , and d^2v/dt^2 at $t=0+$ (4)
- A)



Course Outcome: "CO2" / Cognitive Level: "Application"

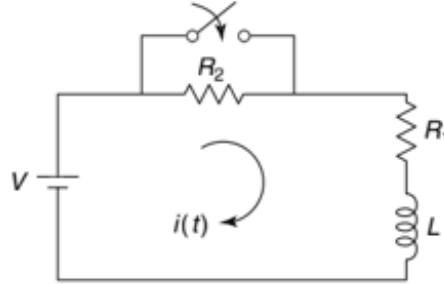
- B) The switch in the figure has been closed for a long time. It is open at $t=0$, Find: (4)
- $i(0^+), v(0^+)$
 - $di(0^+)/dt$ and $dv(0^+)/dt$
 - $i(\infty), v(\infty)$



Course Outcome: "CO2" / Cognitive Level: "Application"

- C) (2)

In the network shown in Figure, the switch is closed at $t = 0$, a steady state having previously been attained. Find the current $i(t)$.

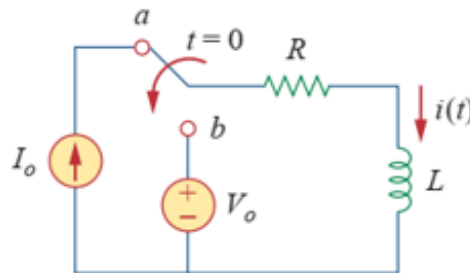


Course Outcome: "CO2" / Cognitive Level: "Application"

3) In the circuit, the switch moves from position a to position b at $t=0$. (4)

Find $i(t)$ for $t > 0$.

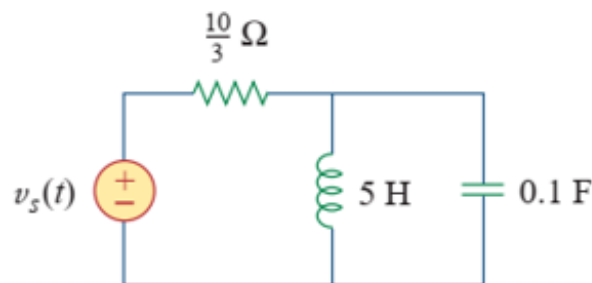
A)



Course Outcome: "CO3" / Cognitive Level: "Application"

B) Consider the circuit in Figure. Find the value of the voltage (4)

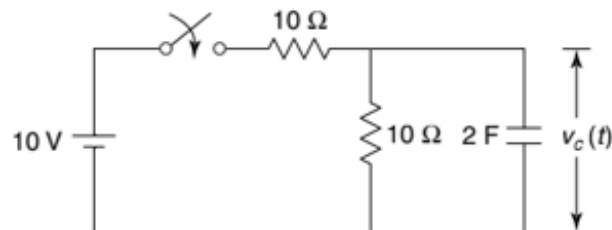
across the capacitor assuming that the value of $v_s(t) = 10u(t)V$ and assume that at $t=0$, $-1A$ flows through the inductor and $+5V$ is across the capacitor.



Course Outcome: "CO3" / Cognitive Level: "Application"

C) The switch in the network shown in the Figure is closed at $t = 0$. (2)

Determine the voltage across the capacitor.



Course Outcome: "CO3" / Cognitive Level: "Application"

4) (4)

- A) Examine whether the following signal is periodic or not? If periodic, determine the fundamental period.

$$\sin(10t + 1) - 2 \cos(5t - 2)$$

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Determine whether the following signals are energy signals or power signals and calculate the energy and power. (4)

$$x(t) = Ae^{-at}u(t), a > 0$$

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Sketch the following signals, $2r(t - 2)$ (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Check the following systems are linear or not. (4)

A) $\frac{dy(t)}{dt} + y(t) = x(t) \frac{dx(t)}{dt}$.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Comment about the linearity, stability, time invariance and causality for the following signal: $y(n) + y(n - 1) = x(n) + x(n - 2)$ (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Find whether the following systems are dynamic or not. (2)

$$y(t) = \frac{d^2x(t)}{dt^2} + 2x(t)$$

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM_2123:Circuits and Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM_2123:Circuits and Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)

2	Understand	0	0(0%)
3	Application	15	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)